CELECOXIB

Sulfonamide allergy: avoid

CV risk

CHEMICAL NAME 4-[5-(4-methylphenyl)-3-(trifluoromethyl)-1H-pyrazol-1-yl]benzenesulfonamide

CLASS Selective COX-2 inhibitor (coxib) NSAID

TRADE NAMES Celebrex · Celcox · Celebra

THERAPEUTIC USES / INDICATIONS

- Osteoarthritis and rheumatoid arthritis.
- Ankylosing spondylitis.
- Acute pain and primary dysmenorrhoea (some countries).
- Patients needing an NSAID with lower GI risk.

PHARMACODYNAMICS

- Selective COX-2 inhibition → ↓ inflammatory prostaglandins.
- Spares COX-1 → less gastric damage and no antiplatelet effect.
- Imbalance favouring thromboxane may ↑ thrombotic risk.

ADMINISTRATION

- **PO** capsules with or without food.
- Capsules may be opened and mixed with applesauce.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dyspepsia, abdominal pain, diarrhoea.
- Headache, dizziness.
- Peripheral oedema, upper respiratory infections.

ADVERSE EFFECTS (SERIOUS)

- Myocardial infarction, stroke.
- GI bleeding (lower than non-selective NSAIDs).
- Renal impairment, hypertension.
- Severe skin reactions; anaphylaxis.

DOSAGES

- **Osteoarthritis:** 200 mg once daily or 100 mg twice daily.
- **Rheumatoid arthritis:** 100–200 mg twice daily.
- **Acute pain:** 400 mg initially, then 200 mg as needed (day 1), then 200 mg twice daily.
- Hepatic impairment (moderate): halve the dose.

CONTRAINDICATIONS & CAUTIONS

- Allergy to sulfonamides.
- Ischaemic heart disease, cerebrovascular disease, heart failure (NYHA II–IV).
- Active peptic ulcer; severe hepatic or renal impairment.
- NSAID-sensitive asthma; third trimester.

KEY INTERACTIONS

- **Warfarin:** ↑ INR.
- **Fluconazole** (CYP2C9 inhibitor): doubles celecoxib levels — halve dose.
- **ACE inhibitors, diuretics, lithium:** as for other NSAIDs.

PHARMACOKINETICS

- **Onset:** 1 h. **Peak:** 3 h.
- Highly protein-bound; metabolised in liver by CYP2C9.
- Excreted in faeces and urine. **t½:** about 11 h.

NURSING MANAGEMENT

- Ask about sulfonamide allergy and heart disease.
- Monitor BP, oedema, chest pain.
- Monitor renal function and LFTs in long-term use.
- Assess pain and joint function.
- Note: does not replace aspirin for cardioprotection.

PATIENT EDUCATION

- Report chest pain, breathlessness, weakness on one side or slurred speech at once.
- Report black stools or swelling.
- Do not exceed the prescribed dose.
- Keep taking low-dose aspirin if prescribed.

KEY POINTS

- Selective COX-2 inhibitor — less GI harm.
- Avoid in sulfonamide allergy and heart disease.
- No antiplatelet effect.
- CYP2C9 metabolism; watch fluconazole.